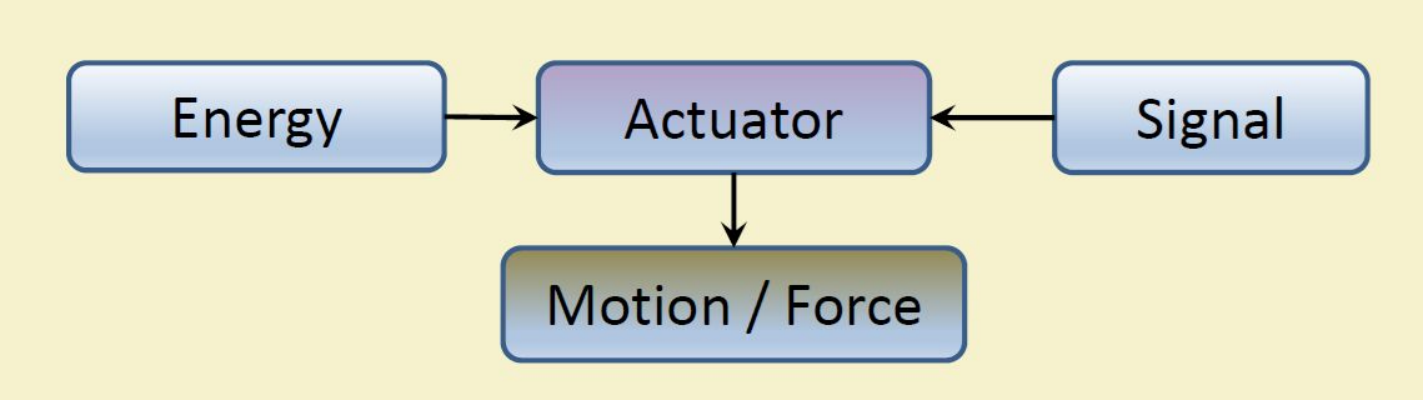
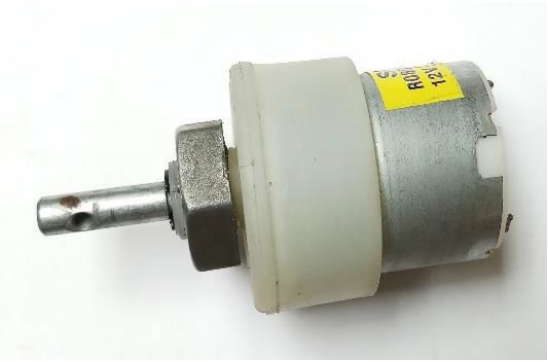
Actuator



- An actuator is a component of a machine or system that moves or controls the mechanism of the system.
- An actuator is the mechanism by which a control system acts upon an environment
- An actuator requires a control signal and a source of energy

Actuator (Contd..)

- After receiving a control signal, the actuator responds by converting the energy into mechanical motion.
- The control system can be simple (a fixed mechanical or electronic system), software-based (e.g. a printer driver, robot control system), a human, or any other input.
- Example: Electric motor, solenoid, hard drive stepper motor, comb drive, hydraulic cylinder, piezoelectric actuator, and pneumatic actuator



Actuator Types

- Hydraulic
- Pneumatic
- Electrical
- Thermal/ Magnetic
- Mechanical

Hydraulic Actuator

- A hydraulic actuator consists of a cylinder or fluid motor that uses hydraulic power to facilitate mechanical operation.
- The mechanical motion is converted to linear, rotary or oscillatory motion.
- Since liquids are nearly impossible to compress, a hydraulic actuator exerts considerable force.
- The actuator’s limited acceleration restricts its usage.

Characteristics

- **High Force Output:** Can generate very large forces with small actuators.
- **Stiff System:** Because liquids are incompressible, response is strong and precise.
- **Limited Acceleration:** Slower compared to pneumatic systems.
- **Heavy & Bulky:** Not suitable for lightweight or portable IoT devices.

Pneumatic Actuator

- A pneumatic actuator converts energy formed by vacuum or compressed air at high pressure into either linear or rotary motion.
- Pneumatic rack and pinion actuators are used for valve controls of water pipes.
- Pneumatic energy quickly responds to starting and stopping signals.
- The power source does not need to be stored in reserve for operation.

Pneumatic Actuator Contd.

- Pneumatic actuators enable large forces to be produced from relatively small pressure changes (e.g., Pneumatic brakes can are very responsive to small changes in pressure applied by the driver).
- It is responsible for converting pressure into force.

Electric Actuator

- An electric actuator is generally powered by a motor that converts electrical energy into mechanical torque.
- The electrical energy is used to actuate equipment such as solenoid valves which control the flow of water in pipes in response to electrical signals.
- Considered as one of the cheapest, cleanest and speedy actuator types available.

Thermal or Magnetic Actuator

- These can be actuated by applying thermal or magnetic energy.
- They tend to be compact, lightweight, economical and with high power density.
- These actuators use shape memory materials (SMMs), such as shape memory alloys (SMAs) or magnetic shape-memory alloys (MSMAs).

Mechanical Actuators

- A mechanical actuator converts rotary motion into linear motion to execute some movement.
- It involves gears, rails, pulleys, chains and other devices to operate.
- Example: rack and pinion.

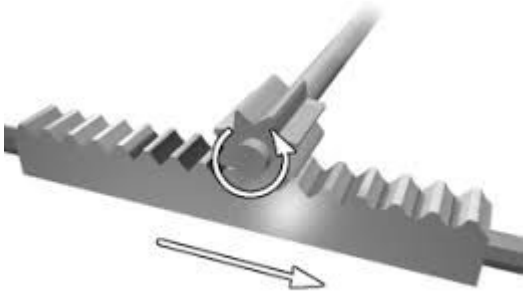


Fig: A rack and pinion mechanism

Soft Actuator

- Soft actuators (e.g. polymer based) are designed to handle fragile objects like fruit harvesting in agriculture or manipulating the internal organs in biomedicine.
- They typically address challenging tasks in robotics.
- Soft actuators produce flexible motion due to the integration of microscopic changes at the molecular level into a macroscopic deformation of the actuator materials.

Shape Memory Polymers

- Shape memory polymer (SMP) actuators function similar to our muscles, even providing a response to a range of stimuli such as light, electrical, magnetic, heat, pH, and moisture changes.
- SMP exhibits surprising features such a low density, high strain recovery, biocompatibility, and biodegradability.

Light Activated Polymers

- Photopolymer/light activated polymers (LAP) are a special type of SMP that are activated by light stimuli.
- The LAP actuators have instant response.
- They can be controlled remotely without any physical contact, only using the variation of light frequency or intensity.

Actuator Characteristics

- After sensors collect data and processors make decisions, actuators carry out the physical action
- Since actuators perform heavy or mechanical tasks, their correct selection is very important for:

Long-term reliability

Operational efficiency

Increased lifetime

Actuator Characteristics

1. Weight:

- Refers to the physical mass of the actuator.
- Heavy actuators are preferred in industrial setups or fixed systems (factories, machinery).
- Lightweight actuators are used in mobile IoT applications like drones, robots, and home automation
- Some heavy actuators are still used in mobile systems → e.g., aircraft landing gears, engine motors.

2. Power Rating:

- Defines the operating power range the actuator can safely handle.
- Usually expressed as power-to-weight ratio.
- Example: Small servo motor (hobby projects): 5 V DC, 500 mA → runs on batteries.
- Industrial servo motor: 460 V AC, 2.5 A → needs a dedicated power supply.
- Exceeding rating = damage or burnout.

Actuator Characteristics Contd.

3. Torque-to-Weight Ratio

•Torque is a rotational force applied by the actuator

•Torque / Weight of actuator

•It Indicates sensitivity and efficiency

- Higher ratio → more torque for less weight (better for mobile robots)
- Lower ratio → heavier actuator, less efficient

4. Stiffness and Compliance

•Stiffness measures resistance to deformation and relates to material elasticity

•Compliance system allows more flexibility (opposite of stiffness)

•Stiff systems → More accurate, faster response (e.g., hydraulic actuators)

•Compliant systems → Flexible, less accurate (e.g., pneumatic actuators)